Software Requirements Specification

for

Bambare Travel Management System

Cross-Platform Mobile Application

Group Name: TMS | Group ID: G01

Client: Bambare Travels (Mr. Dammika Bambaradeniya)

Version 1.0

Prepared by

TMS

Member Name	Registration Number
Avishka Bambaradeniya	SA24100931
Prarthana Meegahakumbura	SA24101440
Shamla Noufer	SA24610780
Oshidi Yapa	SA24610566
P. Vidhuja	SA24101945

Table of Contents

LIST OG TABLE 4

LIST OF FIGURES 5

REVISION HISTORY 6

Project Verification 7

1 INTRODUCTION 8

1.1 Purpose 8

1.2 Intended Audience and Reading Suggestions 8

1.3 Product Scope 9

2 OVERALL DESCRIPTION 10

2.1 Product Perspective 10

2.2 Product Functions 10

2.3 User Classes and Characteristics 11

2.4 Operating Environment 12

2.5 Design and Implementation Constraints 13

2.6 User Documentation 13

2.7 Assumptions and Dependencies 14

2.8 User Interfaces 15

2.8.1 Customer register page 15

2.8.2 Invoice Data Filling Interface 15

2.9 Hardware Interfaces 16

2.10 Software Interface 16

2.11 Communications Interfaces 17

3 System Functional Features 18

3.1 Database Requirements 18

3.2 User Registration & Authentication 21

3.3 Tour Package Browsing & Booking 23

3.4 Map Navigation & Places Recommendation 21
3.5 Expense tracking & Currency Conversion 18
3.6 Admin Dashboard & Reporting 21
4 NON-FUNCTIONAL REQUIREMENTS 27
4.1 Performance Requirements 27
4.2 Safety Requirements 28
4.3 Security Requirements 28
4.4 Software Quality Attributes 29
4.5 Business Rules 30
5 References 32
6 OTHER REQUIREMENTS 33

LIST OF TABLES

Table 2.3.1: User Classes and Characteristics.

Table 2.4.1: Operating Environment — Client-Side Requirements

Table 2.4.2: Operating Environment — Server-Side Requirements

Table 2.10.1: Software Interfaces

Table 3.1.1: Database Entity Summary

Table 3.2.1: Use Case Description UC-01: User Registration & Authentication

Table 3.3.1: Use Case Description — UC-02: Tour Package Browsing & Booking

Table 3.4.1: Use Case Description — UC-03: Map Navigation & Places Recommendation

Table 3.5.1: Use Case Description — UC-04: Expense Tracking & Currency Conversion

Table 3.6.1: Use Case Description — UC-05: Admin Dashboard & Reporting

Table 4.1.1: Performance Requirements

Table 4.4.1: Software Quality Attributes

Table 5.1.1: Revision History

Table 5.2.1: Roles and Responsibilities

LIST OF FIGURES

Figure 2.1.1: System Architecture Diagram (AS-IS to TO-BE)

Figure 2.2.1: High-Level System Overview Diagram

Figure 2.3.1: Use Case Diagram — Overall System

Figure 3.1.1: Entity-Relationship Diagram (ER Diagram)

Figure 3.1.2: Detailed Entity-Relationship Diagram

Revision History

Name	Date	Reason for Changes	Version
G01 TMS Group	10 February 2026	Initial Draft — Project Proposal completed	0.1
G01 TMS Group	18 February 2026	SRS First Draft — Requirements gathered and documented	0.9
G01 TMS Group	27 February 2026	SRS Final Version — Reviewed and approved	1.0

Project Verification

This Software Requirements Specification (SRS) document for the Bambare Travel Management System has been reviewed and approved by the client, Mr. Dammika Bambaradeniya, Director of Bambare Travels. The client's approval confirms that the requirements outlined in this document accurately reflect the business needs and objectives of the organization.

Client Approval	Project Supervisor Approval
Name: Mr. Dammika Bambaradeniya Title: Director, Bambare Travels Signature: _____ Date: _____	Name: Ms. Shanika / Ms. Anjali Title: Project Supervisor, SCU Signature: _____ Date: _____

1 INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document specifies the complete functional and non-functional requirements for the Bambare Travel Management System (TMS), Version 1.0. The system is a cross-platform mobile application designed to digitize and automate the tourism and travel operations of Bambare Travels, a leading Sri Lankan tourism service provider established in 2015.

This document serves as the authoritative reference for all stakeholders involved in the development, testing, and deployment of the system. It defines what the system must do without prescribing how it should be implemented. The SRS covers the complete system scope including user management, tour booking, map navigation, expense tracking, and administrative reporting modules.

1.2 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

- Development Team: Refer to Sections 2, 3, and 4 for technical requirements, system architecture, and functional specifications. Section 3 provides detailed use case descriptions for each module.
- Project Supervisors (Ms. Shanika & Ms. Anjali): Review all sections for completeness, particularly Section 1 (scope), Section 3 (functional features), and Section 4 (non-functional requirements).
- Client (Mr. Dammika Bambaradeniya): Focus on Sections 1.3, 2.2, and 2.3 for product scope and system functions. Section 3 provides use case flows that reflect business processes.
- Quality Assurance Team: Sections 3 and 4 define acceptance criteria, performance benchmarks, and security requirements used for test planning.
- Documentation Writers: Use this document as the basis for user manuals and training materials referenced in Section 2.6.

New readers are advised to begin with Section 1 for project context, proceed to Section 2 for an overall system description, then Section 3 for detailed functional requirements, and Section 4 for quality constraints.

1.3 Product Scope

The Bambare Travel Management System is a comprehensive, cross-platform mobile application targeting Android and iOS devices. The system will serve as the central digital hub for managing all aspects of Bambare Travels' tourism operations, replacing the current manual, paper-based processes.

The system's primary goals are to:

- Automate the booking and reservation process, targeting a 60% reduction in processing time.
- Centralize customer data management for personalized service delivery.
- Provide real-time resource tracking for vehicles, guides, and accommodations.
- Automate financial processes including invoice generation and payment tracking.
- Deliver business intelligence through analytics and reporting dashboards.
- Increase online bookings by 40% within six months of deployment.

The product scope is limited to the mobile application and its associated backend services. Physical infrastructure, third-party payment processing compliance, and destination management beyond the application are outside the scope of this document.

2 OVERALL DESCRIPTION

2.1 Product Perspective

The Bambare Travel Management System is a new, self-contained digital product that replaces the current manual operating system at Bambare Travels. The organization currently relies on phone calls, emails, physical registers, and disconnected spreadsheets for all aspects of their operations. This has resulted in data silos, scheduling conflicts, billing delays, and limited business visibility.

The proposed system implements a modern mobile-first architecture as follows:

- Mobile Client Layer: A Flutter application providing a unified interface for customers and administrators across Android and iOS platforms.
- Authentication Layer: Firebase Authentication for secure, role-based access control.
- Application Services Layer: Modular backend services (Node.js/Express.js) covering booking, customer management, resource allocation, and payment processing.
- Database Layer: Firebase Firestore for real-time synchronization and MongoDB for structured relational data storage.
- External Integrations: Google Maps API for navigation, Currency Conversion API, SMS/Email notification services, and Stripe/PayPal for payment processing.

Figure 2.1.1 illustrates the high-level system architecture showing the transition from the AS-IS manual system to the TO-BE digital system.

2.2 Product Functions

The following summarizes the major functions the Bambare Travel Management System must perform, organized by system module. Detailed specifications are provided in Section 3.

FR1 — User Management & Authentication

- User registration with email verification and Firebase Authentication
- Secure login with role-based access control (Admin / Customer)

- User profile management (CRUD operations)
- Password reset and account recovery mechanisms

FR2 — Tour & Activity Planning and Booking

- Browse, search, and filter tour packages with advanced filtering
- Real-time availability checking and calendar integration
- Customizable tour package creation with pricing options
- Online booking with instant confirmation and booking history management

FR3 — Map, Navigation & Places Recommendation

- Google Maps API integration for location services and navigation
- Interactive map with location pinning and route planning
- Recommended places database with full CRUD operations
- Location-based recommendations for popular destinations and attractions

FR4 — Expense Tracking & Currency Converter

- Personal expense tracking for travelers with categorization
- Real-time currency conversion using external API
- Monthly expense reports, summaries, and budget planning tools

FR5 — Admin Panel, Notifications & Reporting

- Comprehensive admin dashboard with real-time analytics
- Manage hotels, tours, and users with full CRUD operations
- Firebase Cloud Messaging (FCM) for push notifications
- Automated email and SMS notifications for bookings and updates
- System-wide reporting covering bookings, users, and revenue

2.3 User Classes and Characteristics

The Bambare Travel Management System serves two primary user classes, distinguished by their roles, technical expertise, and access privileges.

User Class	Description	Technical Level	Access Level
Customer	Domestic and international tourists using	Basic to moderate	Browse tours, make bookings, track expenses, view maps, receive notifications

	Bambare services	smartphone proficiency	
Administrator	Bambare Travels staff managing system operations and content	Moderate technical proficiency; trained on system	Full CRUD on tours, hotels, users; view reports; send notifications; manage system settings

Table 2.3.1: User Classes and Characteristics

2.4 Operating Environment

The system must operate within the following hardware and software environments:

Client-Side Requirements

Requirement	Specification
Mobile OS (Android)	Android 8.0 (API Level 26) or higher
Mobile OS (iOS)	iOS 12.0 or higher
RAM	Minimum 2GB; Recommended 4GB or higher
Storage	Minimum 100MB available space
Network	Stable internet connection (3G / 4G / WiFi)
GPS	GPS capability required for location-based features
Screen Resolution	720p minimum; 1080p recommended

Table 2.4.1: Operating Environment — Client-Side Requirements

Server-Side Requirements

Requirement	Specification
Cloud Provider	Google Cloud Platform (GCP) with auto-scaling enabled
Server Specifications	4 vCPUs, 16GB RAM (scalable based on demand)
Primary Database	Firebase Firestore (real-time synchronization)
Secondary Database	MongoDB Atlas (structured data storage)
Storage	Google Cloud Storage for media files and backups
Backup Policy	Daily automated backups with 30-day retention period

Table 2.4.2: Operating Environment — Server-Side Requirements

2.5 Design and Implementation Constraints

- Cross-Platform Requirement: The application must function identically on Android 8.0+ and iOS 12.0+, necessitating the use of Flutter for code reusability.
- Internet Connectivity: The system requires an active internet connection for all core features. Offline functionality is limited to cached data viewing only.
- Firebase Dependency: Firebase Authentication and Firestore are mandatory components, binding the system to Google's service availability and pricing model.
- PCI DSS Compliance: Payment processing integration must comply with Payment Card Industry Data Security Standards (PCI DSS).
- Data Privacy: The system must comply with applicable data protection regulations concerning the storage and processing of personal user data.
- Language Support: The user interface must support English and Sinhala as minimum language requirements.
- WCAG 2.1 Level AA: All user interfaces must meet Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standards.

2.6 User Documentation

The following documentation will be delivered alongside the software:

- Customer User Manual: A guide for end-users covering registration, booking tours, using the expense tracker, and navigating maps.
- Administrator User Manual: A comprehensive guide for Bambare Travels staff covering all admin panel functions, reporting, and user management.
- Video Tutorial Series: Step-by-step video guides for both customer and administrator workflows.
- In-App Help: Contextual help tooltips and FAQ section embedded within the application.
- API Documentation: Technical documentation for all backend API endpoints using Swagger/OpenAPI specification.
- Deployment Guide: Technical documentation for system setup, configuration, and production deployment.

2.7 Assumptions and Dependencies

Assumptions

- Users have access to a compatible smartphone running Android 8.0+ or iOS 12.0+ with a stable internet connection.
- The client (Bambare Travels) will provide accurate tour package information, pricing data, and hotel listings for the initial system population.
- Google Maps API will remain commercially available with pricing compatible with the project budget throughout the development lifecycle.
- Firebase services (Authentication, Firestore, Cloud Messaging) will maintain their current API contracts and service availability levels.
- The selected payment gateway (Stripe or PayPal) will be approved for use in Sri Lanka at the time of deployment.

Dependencies

- Google Cloud Platform — Firebase services (Authentication, Firestore, Cloud Messaging, Cloud Storage)
- MongoDB Atlas — Cloud-hosted MongoDB database service
- Google Maps Platform — Maps SDK for mobile, Places API, Directions API
- Currency Conversion API (ExchangeRate-API or similar) — Real-time currency exchange rates
- Stripe or PayPal — Payment gateway services
- SMS/Email Service (Twilio or SendGrid) — Notification delivery
- Flutter (Google) — Cross-platform mobile development framework

2.8 User Interfaces

The application follows Material Design guidelines for Android and Apple Human Interface Guidelines for iOS components. The UI must be intuitive, minimal, and accessible. The following describes the primary interfaces of the system.

2.8.1 Customer Home & Tour Browsing Interface

The customer-facing home screen provides a clean dashboard displaying featured tour packages, a search bar with filter options (price range, destination, date, duration), quick access to active bookings, and a navigation bottom bar with sections for Home,

Bookings, Maps, Expenses, and Profile. Each tour card displays the destination image, tour name, price, duration, and availability status. The interface must load within 3 seconds and handle graceful degradation on slow network connections.

2.8.2 Booking & Confirmation Interface

The booking interface presents a step-by-step wizard: (1) Select dates and group size, (2) Review tour details and pricing, (3) Enter traveler information, (4) Payment processing, (5) Booking confirmation with reference number. The confirmation screen displays a booking summary and triggers an automated email/SMS notification to the customer.

2.8.3 Admin Dashboard Interface

The administrator dashboard provides a real-time overview of system metrics including total bookings, revenue, active tours, and registered users. It includes quick-action cards for common tasks (add tour, manage users, view reports) and graphical charts showing booking trends, revenue by month, and popular destinations. The dashboard is optimized for tablet use but remains functional on mobile screens.

2.9 Hardware Interfaces

The system requires the following hardware interface capabilities from the client device:

- GPS/Location Services: Required for the Map, Navigation & Places Recommendation module (FR3). The application will request location permissions following Android and iOS best practices.
- Camera: Optional access for profile photo upload and document scanning features.
- Storage: Read/write access to device storage for caching tour images and downloaded content for offline viewing.
- Network Interface: The application communicates with all backend services exclusively over HTTPS using the device's WiFi or cellular radio.

2.10 Software Interface

The system interfaces with the following external software components:

Software / API	Version	Purpose
Firebase Authentication	Latest	Secure user login, registration, and session management
Firebase Firestore	Latest	Real-time NoSQL database for dynamic data synchronization
Firebase Cloud Messaging	Latest	Push notification delivery to mobile devices
MongoDB Atlas	6.0+	Structured data storage for bookings, payments, and reports
Google Maps SDK	Latest	Mapping, navigation, location search, and route planning
Currency Exchange API	v6	Real-time currency conversion rates
Stripe / PayPal SDK	Latest	Secure payment processing and transaction management
Twilio / SendGrid	Latest	SMS and email notification delivery
Node.js	18 LTS	Backend runtime for API services
Express.js	4.x	RESTful API framework
Flutter	3.x (Stable)	Cross-platform mobile development framework (Dart)

Table 2.10.1: Software Interfaces

2.11 Communications Interfaces

- HTTPS/TLS 1.3: All communications between the mobile client and backend services are encrypted using HTTPS with TLS 1.3. No unencrypted HTTP connections are permitted.
- RESTful API: The mobile application communicates with backend services via REST API calls using JSON-formatted request and response payloads.
- Firebase Real-Time Listeners: Firestore WebSocket connections are used for real-time data synchronization, such as booking status updates and admin notifications.
- Push Notifications: Firebase Cloud Messaging (FCM) protocol is used for delivering push notifications to Android and iOS devices.

- Email: SMTP protocol via SendGrid or equivalent for transactional emails (booking confirmations, password resets, promotional communications).
- SMS: Twilio API via HTTPS REST calls for SMS notifications for booking confirmations and OTP delivery.
- Payment Gateway: PCI DSS-compliant HTTPS connections to Stripe or PayPal API endpoints for all payment transactions.

3 SYSTEM FUNCTIONAL FEATURES

3.1 Database Requirements

The Bambare Travel Management System database design follows the Third Normal Form (3NF) normalization principles to ensure data integrity, minimize redundancy, and optimize query performance. The system uses a hybrid database approach: Firebase Firestore for real-time synchronization and MongoDB for structured relational data storage.

The database consists of 16 entities connected by 18 relationships. The key entities are: USER, CUSTOMER, ADMIN, TOUR_PACKAGE, BOOKING, PAYMENT, HOTEL, VEHICLE, TOUR_GUIDE, EXPENSE, NOTIFICATION, REVIEW, ITINERARY, DESTINATION, CURRENCY_RATE, and ACTIVITY.

Entity	Type	Description
USER	Strong	Core authentication entity storing login credentials and role for all system users
CUSTOMER	Strong	Extended profile for customer users with travel preferences and history
ADMIN	Strong	Administrator profile with system access permissions
TOUR_PACKAGE	Strong	Tour product catalog with details, pricing, availability, and itinerary
BOOKING	Strong	Reservation records linking customers to tour packages with status tracking
PAYMENT	Weak	Payment transaction records linked to bookings (1:1 relationship)
HOTEL	Strong	Accommodation details, pricing, and availability data
VEHICLE	Strong	Vehicle fleet inventory for transportation management
TOUR_GUIDE	Strong	Guide profiles, specializations, and availability scheduling
EXPENSE	Weak	Customer expense records for personal travel budget tracking

NOTIFICATION	Weak	System notification records for push, email, and SMS communications
REVIEW	Weak	Customer reviews and ratings for tours and accommodations
ITINERARY	Strong	Detailed day-by-day schedule for tour packages
DESTINATION	Strong	Location master data for tour destinations and attractions
CURRENCY_RATE	Strong	Cached exchange rate data for currency conversion feature
ACTIVITY	Strong	Individual activities and experiences within tour packages

Table 3.1.1: Database Entity Summary

The Entity-Relationship diagram illustrating the complete database schema is presented below:

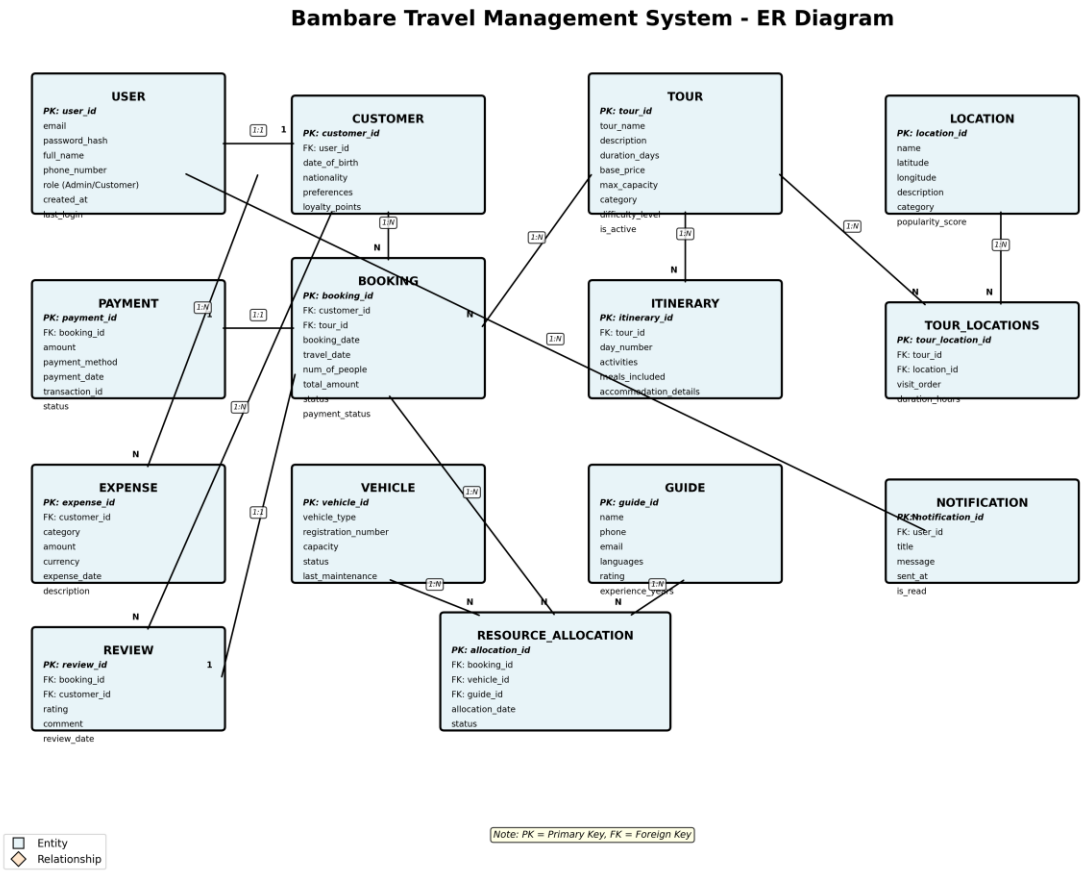

Figure 3.1.1: Entity-Relationship Diagram — Bambare Travel Management System

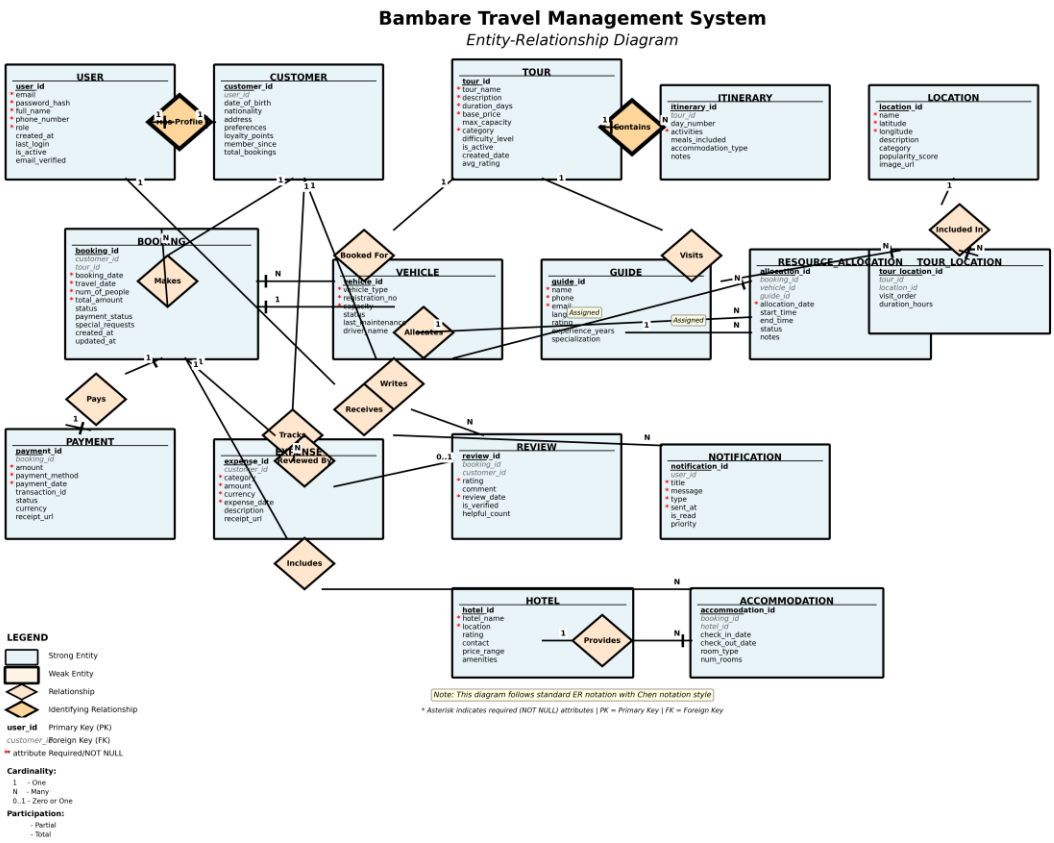

Figure 3.1.2: Detailed Entity-Relationship Diagram

3.2 UC-01: User Registration & Authentication

This use case describes the process by which new users register for the Bambare TMS application and existing users securely log in to access system features. Firebase Authentication handles all credential verification, token management, and session control.

Table 3.2.1: Use Case Description — UC-01: User Registration & Authentication

Use Case ID	UC-01
Use Case Name	User Registration & Authentication
Actors	Guest User (Registration), Registered User (Login)
Pre-Conditions	Application is installed and running. User has a valid email address. Internet connection is available.
Post Conditions	User account is created and verified (Registration). User is authenticated and directed to the home screen with role-appropriate access (Login).

	Action	System Response
Success Path	1. User selects 'Register' and enters full name, email, phone, and password. 2. User submits the registration form. 3. User clicks the email verification link. 4. User logs in with credentials.	1. System validates input fields in real-time. 2. Firebase creates user account; verification email is sent. 3. System marks email as verified and activates account. 4. System authenticates user and redirects to home dashboard.
Alternate Path	User selects 'Forgot Password' and enters registered email address.	System sends a password reset link via Firebase Authentication. User resets password and logs in successfully.
Exception Path	User enters an already-registered email address during registration. User enters incorrect login credentials.	System displays 'Email already in use' error and prompts login. System shows error message; after 5 failed attempts, account is temporarily locked for 15 minutes.
Special Requirements	Password must be minimum 8 characters with at least one uppercase letter, one number, and one special character. Session tokens expire after 24 hours of inactivity.	

3.3 UC-02: Tour Package Browsing & Booking

This use case describes how customers browse available tour packages with filtering options and complete a booking transaction through the system. The booking process includes real-time availability checking and automated confirmation.

Table 3.3.1: Use Case Description — UC-02: Tour Package Browsing & Booking

Use Case ID	UC-02
Use Case Name	Tour Package Browsing & Booking
Actors	Customer (Primary), Payment Gateway (Secondary), Admin (Notification Recipient)
Pre-Conditions	User is logged in as Customer. At least one tour package exists in the system. Payment gateway is operational. Internet connection is available.
Post Conditions	Booking is recorded in the database with CONFIRMED status. Customer receives confirmation notification. Admin receives new booking notification.

	Action	System Response
Success Path	1. Customer opens Tours section and applies filters (destination, date, price). 2. Customer selects a tour package and views details. 3. Customer selects travel date and group size, then clicks 'Book Now'. 4. Customer reviews booking summary and proceeds to payment. 5. Customer completes payment transaction.	1. System displays filtered tour list with real-time availability indicators. 2. System displays full tour details, itinerary, pricing, and availability calendar. 3. System validates availability and calculates total price including taxes. 4. System presents payment gateway integration. 5. System confirms payment, creates booking record, sends confirmation email/SMS, and notifies admin.
Alternate Path	Customer selects a tour but chosen dates are no longer available after selection.	System displays real-time availability update and suggests alternative available dates.
Exception Path	Payment transaction fails due to insufficient funds or gateway timeout.	System cancels the pending booking, releases the reserved slot, and displays a payment error message with retry option. Booking record is set to FAILED status.
Special Requirements	Bookings must be made at least 48 hours in advance. Cancellation policy: free cancellation up to 7 days before tour date; 50% refund for 3-7 days; no refund within 3 days.	

3.4 UC-03: Map Navigation & Places Recommendation

This use case describes how customers use the integrated map feature to explore destinations, find recommended places, and get navigation assistance during their travels in Sri Lanka.

Table 3.4.1: Use Case Description — UC-03: Map Navigation & Places Recommendation

Use Case ID	UC-03
Use Case Name	Map Navigation & Places Recommendation
Actors	Customer (Primary), Google Maps API (Secondary)
Pre-Conditions	User is logged in. Device GPS is enabled and location permission is granted. Google Maps API is available. Internet connection is available.

Post Conditions	Customer has viewed location-based recommendations and/or received navigation directions to a selected destination.	
	Action	**System Response**
Success Path	1. Customer opens the Maps section of the application. 2. Customer views their current location and nearby recommended places. 3. Customer selects a place of interest from the list or map pin. 4. Customer requests navigation directions to the selected location.	1. System loads Google Maps centered on the customer's current GPS location. 2. System displays nearby recommended places as pins on the map with category filters (Beach, Temple, Tea Estate, Wildlife). 3. System shows place details including photos, description, operating hours, and ratings. 4. System launches turn-by-turn navigation via Google Maps Directions API.
Alternate Path	Customer searches for a specific destination by name.	System uses Google Places Autocomplete to provide suggestions as the customer types, then navigates to the selected location on the map.
Exception Path	GPS signal is unavailable or location permission is denied.	System displays a message explaining that location services are required, with a prompt to enable GPS and grant permissions. Map loads with default center point (Sri Lanka).
Special Requirements	Location data is only used during active app sessions and is never stored on servers without explicit user consent.	

3.5 UC-04: Expense Tracking & Currency Conversion

This use case describes how customers track their travel expenses and convert currencies during their trips, enabling better budget management and financial planning.

Table 3.5.1: Use Case Description — UC-04: Expense Tracking & Currency Conversion

Use Case ID	UC-04
Use Case Name	Expense Tracking & Currency Conversion

Actors	Customer (Primary), Currency Exchange API (Secondary)	
Pre-Conditions	User is logged in as Customer. Internet connection is available for currency conversion (expense logging works offline).	
Post Conditions	Expense entry is saved to Firestore. Currency conversion result is displayed. Monthly expense report is available for viewing.	
	Action	**System Response**
Success Path	1. Customer opens Expense Tracker and selects 'Add Expense'. 2. Customer enters amount, category, date, and optional note. 3. Customer saves the expense entry. 4. Customer uses Currency Converter tool to convert LKR to foreign currency.	1. System displays existing expenses categorized by type with total spending overview. 2. System provides a form with category options (Food, Transport, Accommodation, Shopping, Activities, Other). 3. System saves the entry to Firestore and updates the total and charts. 4. System fetches real-time exchange rate from Currency API and displays the converted amount with rate timestamp.
Alternate Path	Customer views monthly expense report.	System generates a visual report showing expense breakdown by category, total spent, and comparison against budget limit (if set). Option to export as PDF.
Exception Path	Currency conversion API is unavailable (network error or service outage).	System uses the last cached exchange rate with a clear indication of the cache date/time, and displays a warning that rates may not be current.
Special Requirements	Expenses are stored per user and are private. Currency rates are cached for a maximum of 4 hours to minimize API calls.	

3.6 UC-05: Admin Dashboard & Reporting

This use case describes how system administrators manage the Bambare Travel Management System through the admin dashboard, including content management, user oversight, and report generation.

Table 3.6.1: Use Case Description — UC-05: Admin Dashboard & Reporting

Use Case ID	UC-05	
Use Case Name	Admin Dashboard & Reporting	
Actors	Administrator (Primary), Firebase Cloud Messaging (Secondary)	
Pre-Conditions	User is logged in with Admin role. Backend services are operational.	
Post Conditions	Administrative action is completed (content updated, user managed, report generated, or notification sent). All actions are logged in the audit trail.	
	Action	**System Response**
Success Path	1. Admin logs in and views the main dashboard with KPI metrics. 2. Admin navigates to tour management and adds a new tour package. 3. Admin views booking reports for the current month. 4. Admin sends a promotional push notification to all customers.	1. Dashboard loads with real-time data: total bookings, revenue, active tours, and registered users. 2. System validates tour details and publishes the new package; it appears in customer search results immediately. 3. System generates a booking report with filtering options by date range, tour, and status. Option to export as PDF/CSV. 4. System uses FCM to deliver the push notification to all subscribed customer devices.
Alternate Path	Admin needs to deactivate a customer account due to policy violation.	Admin locates the user in the Users management section, changes status to Inactive. System revokes active sessions and prevents future logins until reactivated.
Exception Path	Admin attempts to delete a tour package that has active future bookings.	System prevents deletion and displays a warning listing the number of active bookings affected. Admin is prompted to cancel bookings first or archive the package instead.
Special Requirements	All admin actions (create, update, delete) are logged with timestamp and admin user ID in a non-deletable audit log. Reports containing personal data must comply with data privacy requirements.	

4 NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

The system must meet the following performance benchmarks to ensure a responsive and reliable user experience across all network conditions:

Performance Criterion	Target	Rationale
Page/Screen Load Time	< 3 seconds	On standard 4G connection under normal operating conditions
Concurrent User Capacity	100+ users	Without performance degradation during peak tourist season
Database Query Response Time	< 2 seconds (95th percentile)	For all standard CRUD operations under normal load
Real-Time Data Synchronization	< 500ms	Firestore listener updates for booking status changes
Currency API Response	< 2 seconds	Including network round-trip to external currency API
Push Notification Delivery	< 5 seconds	From trigger event to device delivery via FCM
Map Loading Time	< 4 seconds	Including Google Maps tiles and place markers on first load
System Uptime	99.5% per month	Maximum permitted downtime: ~3.6 hours per month

Table 4.1.1: Performance Requirements

4.2 Safety Requirements

The system must implement the following safety measures to protect users and their data from potential loss or harm:

- Data Backup: Automated daily backups of all database content (Firestore and MongoDB) must be performed. Backups must be stored in geographically separate locations with a minimum 30-day retention period.
- Disaster Recovery: A documented disaster recovery plan must be in place with a maximum Recovery Time Objective (RTO) of 24 hours and a Recovery Point Objective (RPO) of 4 hours.

- Transaction Integrity: All booking and payment transactions must use atomic operations to prevent partial failures that could result in data inconsistency (e.g., payment recorded but booking not created).
- Input Validation: All user inputs must be validated both client-side and server-side to prevent injection attacks and ensure data integrity.
- Graceful Degradation: The system must fail gracefully when external services (Maps API, Currency API, Payment Gateway) are unavailable, providing meaningful error messages without crashing.
- Rate Limiting: API endpoints must implement rate limiting to prevent denial-of-service conditions, particularly for the payment processing and authentication endpoints.

4.3 Security Requirements

- Transport Encryption: All data transmitted between the mobile client and backend services must be encrypted using HTTPS/TLS 1.3. No plain HTTP connections are permitted for any API endpoint.
- Authentication Security: User passwords must be hashed using Firebase Authentication's built-in bcrypt-based hashing. The application must never store plaintext passwords.
- Authorization: Role-based access control (RBAC) must be strictly enforced at both the API and Firestore security rules levels. Customer accounts must not be able to access admin endpoints.
- PCI DSS Compliance: The application must not store, process, or transmit raw payment card data. All payment processing must be delegated to the PCI DSS-certified payment gateway (Stripe or PayPal).
- Session Management: JWT tokens must expire after 24 hours of inactivity. Refresh tokens must be securely stored in device secure storage (Keychain on iOS, Keystore on Android).
- Data Privacy: Personally Identifiable Information (PII) must be stored encrypted at rest. Location data must only be collected with explicit user consent and must not be persisted on the server without user permission.
- Penetration Testing: A security audit and penetration test must be conducted before the production launch to identify and remediate vulnerabilities.

- Account Lockout: After 5 consecutive failed login attempts, the account must be temporarily locked for 15 minutes to prevent brute-force attacks.

4.4 Software Quality Attributes

Quality Attribute	Description & Target
Usability	The interface must be learnable within 30 minutes by a non-technical user. All primary user flows (booking, expense tracking, map navigation) must be completable in fewer than 5 taps/steps. WCAG 2.1 Level AA accessibility compliance is mandatory.
Reliability	The system must achieve 99.5% uptime per month. All data operations must maintain ACID properties where applicable. Firestore offline persistence must ensure data is not lost during temporary connectivity loss.
Maintainability	Code coverage for unit tests must be minimum 80% (Flutter Test framework). The codebase must follow defined coding standards and include inline documentation. New features must be deployable without system downtime using rolling deployment strategies.
Portability	The application must run on Android 8.0+ and iOS 12.0+ without platform-specific code exceeding 15% of the total codebase. The backend must be deployable to any cloud provider supporting Node.js with minimal configuration changes.
Scalability	The system architecture must support horizontal scaling to handle a 10x increase in concurrent users during peak tourist season without architectural changes.
Interoperability	All external API integrations must use versioned endpoints with fallback mechanisms. Data exchange must use JSON format following OpenAPI 3.0 specifications.
Testability	All business logic must be implemented in pure functions with no side effects where possible, enabling isolated unit testing. Integration tests must be automated and executable in a CI/CD pipeline.

Table 4.4.1: Software Quality Attributes

4.5 Business Rules

- Booking Lead Time: Customers must complete bookings a minimum of 48 hours before the scheduled tour departure date. Same-day bookings are not permitted.

- Cancellation Policy: Cancellations made more than 7 days before departure receive a full refund. Cancellations between 3 and 7 days receive a 50% refund. Cancellations within 3 days of departure are non-refundable.
- Group Size Limits: Each tour package must define a minimum and maximum group size. Bookings that exceed the maximum capacity must be rejected with an appropriate notification to the customer.
- Role Separation: Administrator accounts cannot be used as customer accounts. A person requiring both admin and customer access must maintain two separate accounts.
- Review Policy: Only customers who have completed a booked tour are permitted to submit reviews for that tour package. Reviews are subject to admin moderation.
- Notification Consent: Promotional push notifications may only be sent to customers who have explicitly opted in to marketing communications during registration or in their profile settings.
- Currency Rate Freshness: Currency exchange rates displayed to users must not be older than 4 hours. The system must clearly display the rate timestamp to the user.
- Admin Audit Trail: All create, update, and delete operations performed by administrators must be recorded in an immutable audit log that cannot be modified or deleted by any system user.

5 REFERENCES

Ref	Citation
[1]	I. Sommerville, Software Engineering, 10th ed. Upper Saddle River, NJ: Pearson, 2015.
[2]	K. Schwaber and J. Sutherland, The Scrum Guide, Scrum.org, 2020. [Online]. Available: https://scrumguides.org
[3]	Firebase Documentation, Firebase for Mobile Development, Google LLC, 2025. [Online]. Available: https://firebase.google.com/docs
[4]	Flutter Documentation, Building Mobile Apps, Google, 2025. [Online]. Available: https://flutter.dev/docs
[5]	Google Cloud Platform, Cloud Services Documentation, Google LLC, 2025. [Online]. Available: https://cloud.google.com/docs
[6]	W3C, Web Content Accessibility Guidelines (WCAG) 2.1, 2018. [Online]. Available: https://www.w3.org/TR/WCAG21/
[7]	MongoDB, Inc., MongoDB Atlas Documentation, 2025. [Online]. Available: https://www.mongodb.com/docs/atlas/
[8]	Stripe, Inc., Stripe API Reference, 2025. [Online]. Available: https://stripe.com/docs/api
[9]	Google Maps Platform, Maps SDK for Android / iOS, Google LLC, 2025. [Online]. Available: https://developers.google.com/maps
[10]	IEEE, IEEE 830-1998 — Recommended Practice for Software Requirements Specifications, IEEE, 1998.

6 OTHER REQUIREMENTS

Appendix A: Glossary

Term / Abbreviation	Definition
TMS	Travel Management System — the software system described in this document
SRS	Software Requirements Specification — this document
CRUD	Create, Read, Update, Delete — the four basic database operations
FCM	Firebase Cloud Messaging — Google's push notification service
FR	Functional Requirement — a specification of system behavior
NFR	Non-Functional Requirement — a specification of system quality attributes
GCP	Google Cloud Platform — the cloud infrastructure provider
API	Application Programming Interface — a set of protocols for software communication
RBAC	Role-Based Access Control — access management based on user roles
JWT	JSON Web Token — a compact, self-contained token for secure information exchange
PII	Personally Identifiable Information — data that can identify a specific individual
WCAG	Web Content Accessibility Guidelines — international web accessibility standards
PCI DSS	Payment Card Industry Data Security Standard — security standards for payment processing
RTO	Recovery Time Objective — maximum tolerable downtime after a system failure
RPO	Recovery Point Objective — maximum acceptable data loss measured in time
3NF	Third Normal Form — a database normalization level minimizing data redundancy

Appendix B: Analysis Models

This appendix references the analysis models developed for the Bambare Travel Management System. All diagrams are stored in the project repository and are available upon request from the project supervisor.

- System Architecture Diagram (AS-IS and TO-BE) — Illustrates the transition from manual operations to the digital platform
- Entity-Relationship Diagram — Complete database schema (included in Section 3.1)
- Detailed ER Diagram — Entity attributes and relationship cardinalities (included in Section 3.1)
- Use Case Diagrams — System-level and module-level use case diagrams
- Data Flow Diagrams (DFD) — Level 0 and Level 1 data flow diagrams
- Class Diagrams — Object-oriented class structure for each system module
- Sequence Diagrams — Interaction flow for key use cases (booking, authentication, payment)
- Activity Diagrams — Business process flows for tour booking and admin operations

Appendix C: To Be Determined (TBD) List

- TBD-01: Final payment gateway selection (Stripe vs. PayPal) — pending merchant account approval for Sri Lanka.
- TBD-02: Specific Currency Exchange API provider and subscription tier — pending cost-benefit analysis.
- TBD-03: Final Sinhala language localization scope — to be confirmed with the client during UAT phase.
- TBD-04: Offline mode functionality extent — architecture decision pending initial performance testing results.
- TBD-05: Third-party review moderation tool integration — to be assessed based on volume of reviews post-launch.

Appendix D: Personal Contributions

The following section documents the individual contributions of each team member to the Bambare Travel Management System project.

Avishka Bambaradeniya — SA24100931

Role: Project Manager & Developer

Assigned Module: Map, Navigation & Places Recommendation Module (FR3)

Key Tasks & Contributions:

- Google Maps API integration for mobile (Flutter Maps / google_maps_flutter package)
- Location pinning and interactive map UI implementation
- Recommended places CRUD operations (Firestore backend)
- Popular destinations database design and data population
- Location-based recommendation algorithm implementation
- Popular places report generation for admin dashboard
- Overall project coordination, timeline management, and stakeholder communication
- SRS document compilation and version control management

Signature: _____

Date: _____

Prarthana Meegahakumbura — SA24101440

Role: UI/UX Designer & Developer

Assigned Module: Expense Tracking & Currency Converter Module (FR4)

Key Tasks & Contributions:

- Expense tracker UI design and Flutter widget implementation
- Expense CRUD operations with Firestore transaction support
- Expense categorization system (Food, Transport, Accommodation, Shopping, Activities, Other)
- Real-time currency conversion integration (Currency Exchange API)
- Monthly expense reports and data visualization (fl_chart package)
- Budget planning and tracking feature implementation
- UI/UX design for customer-facing screens (Figma prototypes)

Signature: _____

Date: _____

Shamla Noufer — SA24610780

Role: UI/UX Designer & Developer

Assigned Module: Tour & Activity Planning and Booking Module (FR2)

Key Tasks & Contributions:

- Tour listing and search UI with advanced filtering implementation
- Tour package CRUD operations (Firestore backend integration)
- Booking CRUD operations with real-time availability validation
- Booking history and management interface
- Automated booking confirmation notification (FCM + Email/SMS)
- Booking status tracking and cancellation workflow
- UI/UX design for booking flow screens (Figma prototypes)

Signature: _____

Date: _____

Oshidi Yapa — SA24610566

Role: Business Analyst & Developer

Assigned Module: Admin Panel, Notifications & Reporting Module (FR5)

Key Tasks & Contributions:

- Admin dashboard UI development with real-time KPI metrics
- Hotel, tour, and user management CRUD operations
- Firebase Cloud Messaging integration for push notifications
- Automated email and SMS notification system
- System-wide reporting module (bookings, users, revenue)
- Database access control rules (Firestore Security Rules)
- Requirements documentation and stakeholder liaison

Signature: _____

Date: _____

P. Vidhuja — SA24101945

Role: Business Analyst & Developer

Assigned Module: User Management & Authentication Module (FR1)

Key Tasks & Contributions:

- Login and registration UI development in Flutter
- Firebase Authentication integration (email/password, email verification)
- User profile CRUD operations
- Role-based access control implementation (Admin/Customer roles)
- Password reset and account recovery workflow
- User registration and activity reports for admin dashboard
- Security requirements documentation and compliance review

Signature: _____

Date: _____